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Anatomy of an American football game: Player to player contact before, during and after an NFL game in context of the 2020 COVID-19 pandemic

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Anatomy of an American football game: Player-to-player contact before, during and after an NFL game in context of the 2020 COVID-19 pandemic

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Abstract:

***Objectives*:** To quantify levels of potential exposure to SARS-CoV-2 surrounding a typical professional American football game, with a focus on interactions on-field between teammates and opposing players before, during, and immediately after competition.

***Methods*:** We examined across-Club consecutive interactions ≥2 minutes within 6 feet [1.8 meters] between athletes on opposing Clubs for all 2020 NFL regular season games (n=256). Cumulative interaction was measured for a representative subset (n=119; 46%) of games. Wearable proximity tracking devices (Kinexon) were used to measure distance and duration of interactions; these data were combined with game schedule and Club rosters for analyses. Frequency and per-game mean, median, interquartile range for consecutive interactions ≥2/≥5 minutes and cumulative interactions ≥5/≥15 were described overall and stratified by pre-game, in-game and post-game.

***Results*:** Of the1,964 distinct player-to-opponent contacts ≥2 minutes in NFL regular season games, the majority (n=1,699; 87%) were fewer than 5 minutes in consecutive length. Among the mean 7.7 distinct contacts ≥2 minutes with opponents each game (median=4; IQR=2, 8), very few were ≥5 consecutive minutes at any point (mean=1.0; median=0; IQR=0, 0). Most (n=849; 43.2%) distinct contacts were pre-game, 546 (27.8%) were during competition, and 569 (29%) were post-game. In games where cumulative interactions were analyzed, there was an average of 17.1 player/opponent interactions with cumulative exposure ≥5 minutes (median=12; IQR=4, 30), almost all of which occurred during competition.

***Conclusion*:** There is limited and short contact between and among competing players in professional American football. In the setting of infectious disease such as the COVID-19 pandemic, a robust prevention program integrating masking, distancing, hygiene and ventilation

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when off-field can be created to minimize on-and-off-field exposures, which effectively reduces transmission risk in outdoors and/or well-ventilated stadium settings.

Keywords: infectious disease, transmission, NFL, COVID-19, SARS-CoV2, contact tracing, wearable technology

Introduction

Respiratory viruses such as SARS-CoV-2 can spread from one person to another through several routes, including respiratory secretions (e.g., droplets or aerosols) and environmental contamination. During the course of the COVID-19 pandemic, SARS-CoV-2 has been predominantly spread via respiratory secretions [1], with transmission via respiratory droplets requiring close contact and prolonged exposure [2]. With these transmission routes in mind and the close contacts required to play American football, the National Football League (NFL) and NFL Players Association (NFLPA) instituted comprehensive safety protocols prior to the 2020 NFL Season. To minimize spread of SARS-CoV-2 [3,4], these protocols were anchored by preventative behavioral protocols, frequent routine Reverse Transcription Polymerase Chain Reaction (RT-PCR) testing and extensive contact tracing.

In contact sports such as American football, players have many interactions and potential contacts on game day with teammates, opponents, and gameday personnel. This applies not only to interactions during the game, but also to contacts before and after a game. 2020 NFL gameday protocols [2] limited the number of people allowed in the locker room and on the field, along with masking on sidelines and limited pre- and post-game interactions between players.

Existing studies of COVID-19 in sport have largely focused on off-field activities [5]. Little data exists on actual exposure time during competition, including both consecutive and cumulative

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[6,7,8,9]. While studies examining potential for spread of COVID-19 in rugby and soccer [7,10,11, 12] to our knowledge, no study has been published measuring the number and duration of interactions using wearable proximity devices, a presumptive proxy for COVID-19 exposure, between players during American football. The objective of this paper is to quantify time of interaction and potential close person-to-person exposure surrounding a typical professional American football game during the COVID-19 pandemic, with a focus on interactions on-field between teammates and opposing players, before, during, and immediately after competition.

Methods:

This analysis examined interactions of close physical proximity (defined below) in all 256 NFL regular season games from September 10, 2020 through January 3, 2021, with focus on interactions both with opponents and among teammates.

Protocols for wearing Kinexon [13] proximity devices (described below) and data were generated to support contact tracing to prevent spread of COVID-19 as part of the NFL's 2020 COVID-19 Health and Safety Program protocols [1,2,14]. Individuals were tested daily or routinely, and full contact tracing was performed for any player or staff with a positive test, providing information on the nature of the interactions and supplemented by Kinexon proximity data . Any positive player or staff, or high-risk close contact of an infection individual, was subsequently quarantined [1].

Data Sources

Game Schedules and Player Characteristics

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Game schedules, start times, locations, and durations, as well as game-by-game Club rosters, were obtained from the NFL Game Stats & Information System (GSIS).

Proximity Tracking

All Club personnel (players, coaches, executives, game day staff) were required to wear Kinexon [13] proximity tracking devices during games and Club activities, including Club travel and any time spent in Club facilities. Each player was assigned two Kinexon devices. The first was a home tag, worn or kept on the player at all times in-facility, traveling with the Club (ground, air transport as well as hotel stays), and up until changing into uniform for a game. The second was a gameplay tag, sewn into a pouch on the player's jersey. These devices collect and store data, including identification information, based on proximity to other devices within a set parameter (in this case 6 feet [1.8m]) [15,16]. The Kinexon devices use a local positioning system (LPS) with distance ranging and operate sampling frequency of 1 Hz. The devices were linked to a roster with unique IDs and names, Club affiliations, and employment tiers [17]. Internal validation of Kinexon devices found a 20cm max deviation in range. The devices did not provide geolocation information about the contact (e.g., sideline, field, locker room).

This raw proximity data was periodically synchronized through central charging stations and aggregated in a centralized database. It was then smoothed to control for measurement and localization error (noise), which can limit the accuracy of the instantaneous distance readings.

A contact was defined as a single interaction between two individuals. Therefore, if two players interacted with each other, this was counted as two contacts (one from each person). Analyses

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are restricted to durations of smoothed contact events; distance-based measures for contact events (e.g., mean distance between pairs of players) are not considered.

The smoothing of raw contact events was accomplished through sequential application of Maximum Distance (1.8 meters), Maximum Sequence Gap (15 seconds), and Minimum Consecutive Duration (120 seconds for consecutive analyses; <1 second for cumulative analyses). One (smoothed) contact event is defined as two devices within the Maximum Distance, with separation for less time than the Maximum Sequence Gap. Smoothed contact event times and durations, once calculated, were restricted to contact events greater than or equal to Minimum Duration.

Inclusion Criteria

All games where the player was on his Club's 53-man roster on that date and designated as a 'Starter', 'Played' (i.e., Backup), 'Did Not Play', or 'Not Active' were included in the study population. Players under contract with a Club but designated as injured reserve or practice squad players for a specific Club-game were not included in that game's analysis, despite proximity data being available for them, as these players were often in locations other than on the field or locker room (e.g., watching the game in a suite). Based on these criteria, a total of 2,286 distinct players who were on a Club's roster for at least one game were included.

In late July 2020, at the start of NFL preseason training camps, the U.S. Centers for Disease Control and Prevention (CDC) defined a close contact as interaction with an infected individual within 6 feet for more than 15 consecutive minutes, which was subsequently updated by the

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CDC to 15 cumulative minutes within a 24-hour period on October 21, 2020 [18]. Throughout the season, interactions of 2 consecutive minutes or longer were used to identify individuals for further investigation during contact tracing, to align both with global guidance on transmission and NFL-based evidence showing that transmission could occur in fewer than 15 minutes [3,8,19,20]. Smoothed single contact events <2 minutes were excluded from consecutive contact analyses for 137 games.

A sensitivity analysis was performed calculating cumulative contact interaction using all contact events regardless of duration (e.g., as short as <1 second) for a subset of 119 games (46%). This subset was selected purposefully, ensuring each Club was represented with data from between 6-9 games', with at least one home game and one away game for each. This subset containing all contact events regardless of time was not available for all games due to data size and constraints; the119 games analyzed compromise a representative sample that encompass all 30 NFL stadiums, varying Club travel schedules, game start times, and months during the NFL regular season. These 119 games contained data on 94% (2,144 of 2,286) of all players because all 30 Clubs were represented, and the majority of NFL players are on their Clubs' rosters for most of the Club's regular season games.

A game time window of 90 minutes prior to game start to 30 minutes post-game was empirically established based on discussions with on-the-ground operations and Club infection control officers; contacts registered by gameplay tags outside of this window were likely due to device transmission after shoulder pad placement but before gameplay and were therefore

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excluded from analyses. Practice, travelling and other contacts outside of game day were excluded from this analysis.

Statistical Analyses

Descriptive statistics including mean, median, and interquartile range (IQR) summarize the frequency of contacts with opponents and teammates that occurred on game days. We analyzed contacts of ≥2 minutes and ≥5 minutes in consecutive length (aligning with on-the-ground contact tracing considerations) and ≥5 minutes and ≥15 minutes in cumulative length (aligning with CDC guidance regarding close contact [2]) and our previous findings that transmission occurs in fewer than 15 minutes of interaction [3]. These were summarized by game and by player.

We analyzed across-Club interactions between players and opponents before (pre-game), during (game time), and immediately after (post-game) competition, to quantify both competition-related and non-competition-related interactions between Clubs. These three segments capture different types of interaction; for example, the pre-game period is often characterized by social interactions, compared to during games when interactions are competitive. To provide a reference point, we analyzed select interactions among players on the same Club to understand the magnitude of teammate-to-teammate interactions and to compare to the magnitude of between-Club interactions. Poisson models were constructed to estimate the crude incidence rate ratio of the number of contacts with opponents versus teammates by game. A chi-squared test was performed to determine the association between positional assignment as an Offensive/Defensive Lineman and proportion of players with at least one interaction ≥5 minutes

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cumulative, as Linemen are potentially more likely to have sustained close contact with opponents due to their roles and typical actions during a game.

Results

Consecutive Contact

The NFL regular season occurred from September 10 through January 3, completing 256 games across 32 Clubs and 2,286 distinct players. In total, there were 1,964 player-to-opposing player contacts ≥2 minutes over all 256 games. 849 (43.2%) of the contacts were pre-game, 546 (27.8%) were during the game, and 569 (29%) were post-game. The mean distinct number of contacts between opposing players ≥2 minutes at any point (pre, during, or post) per game was 7.7 (median=4; IQR=2, 8); on average, 3.3 contacts between opponents occurred before the game (0; 0, 2), 2.1 occurred during the game (0; 0, 2), and 2.2 occurred post-game (0; 0, 4) (Table 1). Among the 2,286 players included in this study, 802 (35%) had one or more contact ≥2 minutes with an opponent at any point in the game (pre, during, or post) over the entire season. In contrast, 100% of players had contact ≥2 minutes with one or more teammates during a game over the entire season, with a mean of 2,914 teammate-to-teammate contacts ≥2 minutes per NFL game during game time (both Clubs), compared to 2.1 player-to-opponent game time contacts per game during game time. The mean number of contacts per game with opponents was 7.7 (95% CI: 5.9 to 10.0). In contrast, the mean number of contacts per game with teammates was 7,559.7 (95% CI: 7,042.0-8,115.5); as these CI's do not overlap, the number of teammates contacts compared to opponent contacts is significantly different.

Of the 256 regular season games, 36 (14%) games had contacts ≥5 consecutive minutes between players on opposing Clubs (mean per game=1.0, median=0; IQR=0, 0). Of the 2,286 players,

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only 123 (5%) had any recorded contacts ≥5 minutes with an opponent over the entire season. The majority of these occurred during pre-game (mean per game=0.9; median=0; IQR= 0, 0); almost no contacts ≥5 minutes occurred during the game (0.1; 0; 0, 0) or post-game (0.05; 0; 0, 0). Same Club player-to-player contacts provide an interesting contrast, with 2,273 of players (99.4%) having an in-game contact ≥5 minutes with a teammate during the season. Overall, there was a mean of 607 teammate-to-teammate in-game contacts ≥5 consecutive minutes, vs. 0.1 between opponents.

Cumulative Contact

Among the 119 games for which cumulative interaction times between player/opponent pairs was examined (Minimum Consecutive Duration of <1 second was used to examine), 2,144 distinct players were included with a total of 160,152 contacts per game. The mean duration of cumulative player-to-opponent by-game exposure was 34.4 seconds (median=14 seconds; IQR=5, 34). Of these, there were 2,030 player-to-opponent by-game contacts with a cumulative duration ≥5 minutes, 396 contacts with a cumulative duration ≥10 minutes, and 119 contacts with a cumulative duration ≥15 minutes (Table 2). On average among these games, 17.1 player/opponent cumulative interactions ≥5 minutes (median=12; IQR=4, 30) occurred per game. The majority of this exposure occurred during the game (mean=15.6; median=12; IQR:2, 26), with less exposure pre-game (0.90; 0; 0, 0) or post-game (0.08; 0; 0, 0). Of the 2,144 players, only 568 (26%) had any recorded contact with an opponent with a cumulative exposure ≥5 minutes. A player had on average 0.15 interactions with opponents with a cumulative exposure ≥5 minutes per game (0; 0, 0).

In the sub-analysis examining in-game exposure by position, 2,042 distinct players designated as a 'Starter' or 'Played' were included across the 119 games. Within this sub-

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population, 1,850 player-to-opponent in-game contacts occurred with cumulative exposure ≥5 minutes. Of those 2,042 players, only 486 (24%) had any recorded contact with an opponent during the game with a cumulative exposure ≥5 minutes, and 372 (76.4% of the 486 with ≥5 minutes of exposure) were by players designated as Linemen in that game. The chi-squared test between positional assignment as a Linemen and any ≥5 minutes cumulative opponent contact was P<.0001.

Discussion

While the monitoring program aimed to detect all COVID-19 cases prior to competition through routine testing and symptom reporting [14], characterizing the in-game unmasked interactions to understand exposure in the setting of an undetected case is important. Despite the high contact nature of an American football game, we found that players do not have sustained time exposure, consecutive or cumulative, with opponents on a football field as measured by the proximity device. Durations of contact between opponents were typically fewer than 2 consecutive minutes; any consecutive contact with an opponent greater than 5 minutes occurred in only 14% (n=36) of regular season games. Game play was of focus here, as in a game, players do not cloth wear masks, are in close contact by nature of the competition, are interacting with individuals outside of their own community, and/or are unable to uphold other preventative measures that were otherwise adhered to in other settings throughout the 2020 season. The purpose of this study was to better understand potential risk of spread in this unique environment. As defined by the CDC on October 21, 2020, close contacts are cumulative interactions 15 minutes or greater [18]. By this definition, close contacts were extremely rare, occurring less than one time per NFL game on average; among the 46% of games examined, less than 5% of players had any

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cumulative interaction with opponents from 90 minutes prior to game start through 30 minutes post-game.

Pre-game interactions between opponents

Under the NFL-NFLPA Protocols, players were required to wear a mask during the pre-game period and maintain physical distancing (of at least 6 feet from other individuals) as much as possible [2]. The Protocols also limited the personnel permitted on-field during pre-game warmups in order to provide sufficient space for physical distancing. That said, player-to-opponent contacts were most common and longest pre-game (0.9 contacts per game ≥5 minutes pre-game compared to 0.05 during the post-game period), likely occurring during pre-game warmups, when both Clubs were on the field and players from opposing Clubs converse with each other. Unlike the post-game period, when players are more focused on discussing the result of the recently concluded game, and the visiting Club is preparing for their imminent return travel, the pre-game period typically affords more unstructured opportunities for between-Club socializing. This setting was well ventilated, and the players were required to follow masking and distancing protocols (noting varying degrees of compliance).

Despite the risks, only 13% of players had more than 2 minutes of consecutive contact with an opponent pre-game, and less than 5% had 5 or more minutes of cumulative exposure in this timeframe. This minimal pre-game contact, likely driven by 2020 NFL COVID-19 protocols, is important, as it could be argued that pre-game activities are more likely to be high-risk exposures in terms of social greetings and conversation. During a game, the limited sustained time of exposure and low cumulative time with opponents, in addition to well-ventilated and often outdoor stadium setting, reflects what was likely a low risk of on-field transmission of SARS-CoV-2 [1].

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Same-Club contacts

Teammates have more sustained contacts throughout a game, especially on the sidelines; targeting behavioral interventions to prevent COVID-19 infection on sidelines (e.g., masking, social distancing) is key to reducing transmission [3]. Of note, there were a mean of 2,914 contacts ≥2 minutes per game between teammates, compared to fewer than 3 of such contacts per game between opponents. Although geolocation data was not available in this study, the sustained contacts between teammates were likely on the sidelines rather than during game play. In the NFL season, sideline mask use for players was recommended in the beginning of the season and then required per protocol beginning Week 12 (November 26), a rule enforced by disciplinary actions and/or fines due to the importance of this finding. Sideline mask use for all non-players with bench area access was required per the protocol for the entirety of the season; these individuals included Club-designated personnel (coaches, training staff, etc.), ownership, game officials, chain crew, ball crew, coach-to-coach technicians, unaffiliated neurotrauma consultants (UNCs) [21], airway management physicians (AMP), visiting team medical liaison (VTML), and instant replay field communication and technicians.

Contacts with opponents during competition

Contacts of ≥2 consecutive minutes between opponents during game play were very rare (2.13 per game), likely due to constant movement on the field. Cumulative contact between opponents during game play of greater than 5 minutes occurred an average of 15.55 times per game, with cumulative 15 in-game minutes occurring very rarely (0.59 times per game) and to only 2.7% of players. This low amount of interaction combined with the high ventilation of NFL stadiums (often outside or open air) renders in-game transmission risk low. This is important, because while behavioral restrictions can be implemented on the sidelines and pre/post competition,

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player to opponent contact during an NFL game is less modifiable and during these unmeasured short contacts, proximity may be very close, often with direct contact and heavy breathing. To mitigate transmission risk, Clubs with domed stadiums were required to submit a Testing, Adjusting and Balancing (TAB) Report in order to ensure proper air flow and filtration throughout the stadium (e.g., heating, ventilation, and air conditioning (HVAC) system tests and balancing focused on air flow/static pressure, details on the amount of outside air introduced into the stadium, number of air changes/hour, and level of Minimum Efficiency Reporting Value (MERV) filtration in place) [2]. The NFL daily testing protocols [2,14]were also a key mitigation measure focused on identifying positive players early in infection, often before viral loads were high, and preventatively quarantining players with known high risk exposure to COVID-19 [1,22] , limiting, though not eliminating, the possibility of on-field transmission.

Post–Game contacts

The fewest number of contacts between opponents occurred post-game, but these were potentially higher risk because these types of contacts often include close interaction such as hugs and handshakes. Players were less likely to be masked because they were still on the field and typically had removed their helmets. Accordingly, post-game interactions between players and staff were limited beginning in Week 12 (November 26). Pursuant to the Protocols, players were required to put on a mask at the conclusion of a game and were then permitted to briefly exchange greetings with the opposing players before proceeding to their respective locker rooms. Additional measures were already in place to reduce contact between players on opposing Clubs after they left the field, such as requiring that all departures from each locker room be staggered unless separate pathways to stadium departure areas were blocked off and used [2].

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The 2020 NFL season was played in the setting of a respiratory virus pandemic, with COVID-19 cases increasing over the course of the regular season [23]. Frequent testing and extensive contact tracing were in place throughout the NFL season, with attention to the Clubs with known exposure at any point in the season. While COVID-19 transmission occurs through respiratory droplets or aerosols and therefore can occur in interactions less than 15 minutes, [1] findings from this study demonstrate that prolonged exposure time between opponents are infrequent during a typical NFL game. Coupled with the outdoor and/or well-ventilated environment [1,24], the risk of COVID-19 transmission during a typical 2020 NFL game was low.

Limitations

This study is limited in that, while time of exposure was measured precisely, it does not measure the intensity or acuity of exposure; instances of aerosol-producing activities (speaking, shouting, spitting) and mucosal contacts were not recorded. Respiratory rate, tidal volume, overall viral load, and other host factors likely drive transmissibility of SARS-CoV-2 [25]. Data from the Kinexon proximity devices do not include geolocation or orientation of the wearer to allow for exact determination of where players are during interactions with each other on the field and whether they were facing each other, especially for interactions within the same Club, which limits the ability to quantify risk per interaction. Cumulative exposure was not available for 137 games due to data availability; however, the 119 games analyzed for cumulative exposure are a representative sample encompassing/including all 30 NFL stadiums, typical Club travel schedules, game start times, months during the NFL regular season, as well as the majority of players. Methods used are not generalizable beyond players to other staff, other levels of American football, or other sports organizations, as the Clubs were operating under strict COVID-19 protocols in the 2020 season, with compliance regulated with potential punitive

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actions. Non-compliance with COVID-19 protocols likely compromises the effectiveness of these measures. Low risk of transmission may not be generalizable to variants such as Delta (B.1.617.2) or Omicron (B.1.1.529) that became predominant in the United States after the conclusion of the 2020 season. This study does not specifically investigate transmission, but rather illustrates contact volume; this is a point for further review and empirical analysis.

Conclusion

In a game of professional American football, there is limited brief, yet intense contact between and among players on the field, especially among opposing players, who provide additional risk by coming from different environments and communities. Interactions usually take place in a well-ventilated indoor stadium or outdoors, leading to minimal on-field transmission risk when combined with protocols that regulate the circumstances of contact between teammates before and after game time (e.g., masking, distancing), such as on sidelines or in the locker room. When a robust off-field program integrating masking, distancing, hygiene and ventilation is established and followed outside of the competition itself, the short length of on-field exposures coupled with the high ventilation of outdoor sports such as American football can limit transmission of respiratory viruses such as SARS-CoV-2. As we move through year 2 and beyond of the COVID-19 pandemic, ongoing monitoring and reassessment of prevention strategies will be required as new variants emerge and spread to further ensure low risk of transmission during outdoor sports activities.

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Table 1. Descriptive statistics: Consecutive player-to-player contacts by game and by player

	Per Game: Mean (Median [IQR]) (N=2,286 players, 256 games)		Per Player: n (%) (N=2,286 players, 256 games)	
	Contacts ≥2 minutes	Players with at least one contact ≥2 minutes	Players with at least one contact ≥2 minutes	Players with at least one contact ≥5 minutes
90 minutes pre-game: Opponent	3.32; 0 [0, 2]	1.76; 0, [0, 2]	287 (13%)	103 (5%)
Game Time: Opponent	2.13; 0 [0, 2]	1.61; 0 [0, 2]	309 (14%)	7 (0.3%)
Game time: Same Club	2,913.6; 3,006 [2,471, 3,299]	97.1; 99 [96, 101]	2,261 (99%)	2,213 (97%)

Post-game (30 minutes after game ends): Opponent	2.22; 0 [0, 4]	2.12; 0, [0, 3.5]	417 (18%)	13 (0.6%)
Overall: Opponent	7.67; 4 [2, 8]	5.23; 4 [2, 6.5]	802 (35%)	123 (5.4%)
Overall: Same Club[a]	7,559.7; 6,577 [5,551, 8,306]	104.0; 104 [102, 106]	2,277 (100%)	2,273 (99%)

[a] Due to the high volume and likely under-measurement due to variance in placement/removal of devices of teammate-to-teammate interaction before and after game, specific pre- and post-game statistics are not shown here and overall numbers may be an underestimate in cases such as device placement and removal in locker room pre- and after game when together with their teammates.

Table 2. Descriptive statistics: Cumulative player-to-opponent exposure by game and by player

	Per Game: Mean (Median [IQR]) (N=2,144 players, 119 games)		Per Player: n (%) (N=2,144 players, 119 games)	
	Player/Opponent interactions with ≥5 minutes cumulative	Player/Opponent interactions with ≥15 minutes cumulative	Players with at least one interaction ≥5 minutes cumulative	Players with at least one interaction ≥15 minutes cumulative
90 minutes pre-game: Opponent	0.90; 0 [0, 0]	0.39; 0 [0, 0]	75 (3.5%)	32 (1.5%)
Game Time: Opponent	15.55; 12 [2, 26]	0.59; 0 [0, 0]	487 (22.7%)	57 (2.7%)
Post-game (30 minutes after game ends): Opponent	0.08; 0 [0, 0]	0.02; 0 [0, 0]	9 (0.4%)	2 (0.1%)
Overall: Opponent	17.06; 12 [4, 30]	1.03; 0 [0, 0]	568 (26.4%)	93 (4.3%)

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